

QUE1 : Execute all the commands explained in this section and write the output.

ANSWER:

##GREP COMMAND :

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cat  
example.txt
```

```
Example list :
```

```
Spain  
France  
England  
Denmark  
Germany  
France  
Germany  
Germany
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep Germany  
example.txt
```

```
Germany  
Germany  
Germany
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep --color  
France example.txt
```

```
France  
France
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep ^D  
example.txt
```

```
Denmark
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep -v  
Germany example.txt
```

```
Example list :  
Spain  
France  
England  
Denmark  
France
```

##SORT COMMAND:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ sort  
example.txt
```

```
Denmark  
England  
Example list :  
France  
France  
Germany  
Germany  
Germany  
Spain
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ sort -r  
example.txt
```

```
Spain  
Germany  
Germany  
Germany  
France  
France
```

```
Example list :
England
Denmark
```

##WC COMMAND:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ wc example.txt
11 11 78 example.txt
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ wc *
 11      11      78 example.txt
1375    7894 419909 Lab 2
-Advanced_UNIX_Shell_Commands_With_Examples.pptx
2377    9885 320919 Lab 2.pdf
 63      102   1098 Lab2_q1.txt
3826   17892 742004 total
```

##CUT COMMAND:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cut -c 1-3
example.txt
Exa
Spa
Fra
Eng
Den
Ger
Fra
Ger
Ger
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cut -c 1-3,6
example.txt
Exal
Spa
Frae
Engn
Denr
Gern
Frae
Gern
Gern
```

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cut -c3-
example.txt
ample list :
ain
ance
gland
nmark
rmany
ance
rmany
rmany
```

#SED COMMAND:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cat
example.txt
Example list :
Spain
France
England
```

Denmark
Germany
France
Germany
Germany

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ sed -e  
's/Germany/Desturchland/' example.txt
```

Example list :
Spain
France
England
Denmark
Desturchland
France
Desturchland
Desturchland

##TR COMMAND:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ cat  
example.txt
```

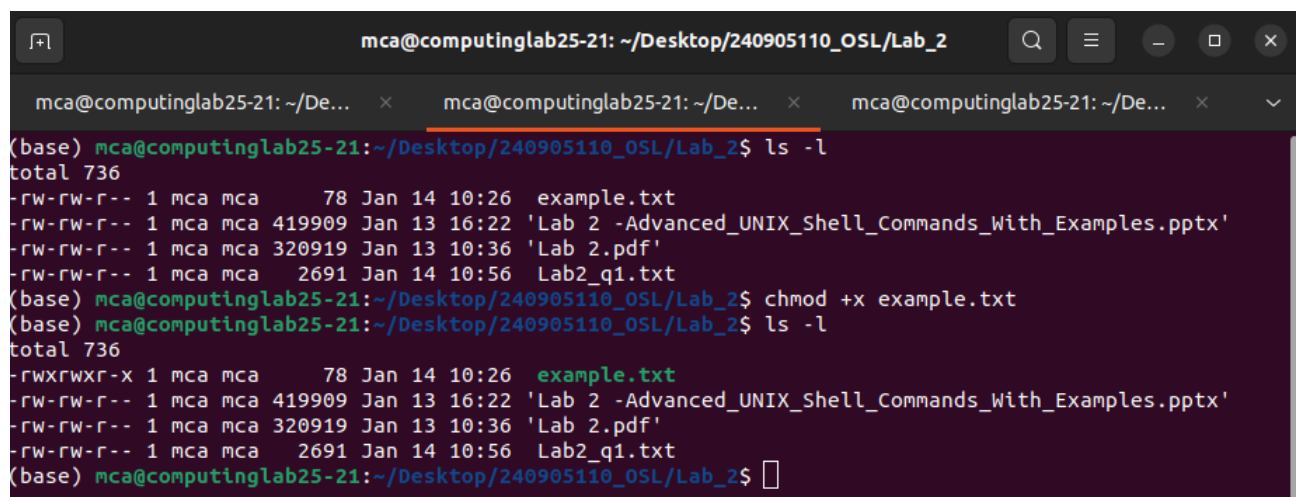
Example list :
Spain
France
England
Denmark
Germany
France
Germany
Germany

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ tr 'a-z' 'A-Z'  
< example.txt
```

EXAMPLE LIST :
SPAIN
FRANCE
ENGLAND
DENMARK
GERMANY
FRANCE
GERMANY
GERMANY

##CHMOD COMMAND:

(example1.txt made into an executable as it turned green in ls -l)



```
mca@computinglab25-21: ~/Desktop/240905110_OSL/Lab_2  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ ls -l  
total 736  
-rw-rw-r-- 1 mca mca 78 Jan 14 10:26 example.txt  
-rw-rw-r-- 1 mca mca 419909 Jan 13 16:22 'Lab 2 -Advanced_UNIX_Shell_Commands_With_Examples.pptx'  
-rw-rw-r-- 1 mca mca 320919 Jan 13 10:36 'Lab 2.pdf'  
-rw-rw-r-- 1 mca mca 2691 Jan 14 10:56 Lab2_q1.txt  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ chmod +x example.txt  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ ls -l  
total 736  
-rwxrwxr-x 1 mca mca 78 Jan 14 10:26 example.txt  
-rw-rw-r-- 1 mca mca 419909 Jan 13 16:22 'Lab 2 -Advanced_UNIX_Shell_Commands_With_Examples.pptx'  
-rw-rw-r-- 1 mca mca 320919 Jan 13 10:36 'Lab 2.pdf'  
-rw-rw-r-- 1 mca mca 2691 Jan 14 10:56 Lab2_q1.txt  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$
```

##KILL COMMAND:



```
mca@computinglab25-21: ~  
(base) mca@computinglab25-21:~$ ps -e | grep mines  
8050 ?          00:00:00 gnome-mines  
(base) mca@computinglab25-21:~$ kill 8050  
(base) mca@computinglab25-21:~$ ps -e | grep mines  
(base) mca@computinglab25-21:~$
```

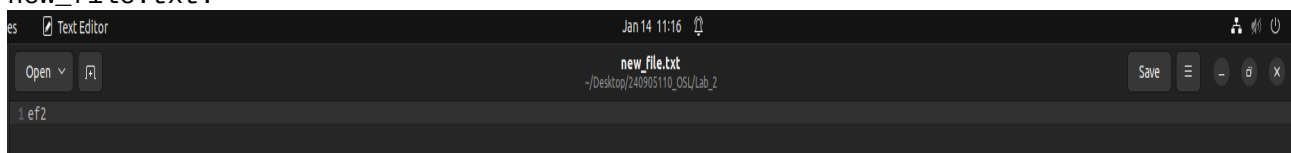
QUE2:2. Write grep commands to do the following activities:

- To select the lines from a file that have exactly two characters.
- To select the lines from a file that start with the upper case letter.
- To select the lines from a file that end with a period.
- To select the lines in a file that has one or more blank spaces.
- To select the lines in a file and di

ANSWER:

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep '^..$'  
example.txt  
ab  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep '^[A-Z]'  
example.txt  
Example list :  
Spain  
France  
England  
Denmark  
Germany  
France  
Germany  
Germany  
United Kingdom  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep '\\.$'  
example.txt  
cd.  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep ' *'  
example.txt  
Example list :  
United Kingdom  
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep '[0-9]'  
example.txt > new_file.txt
```

new_file.txt:



```
new_file.txt  
1 ef2
```

QUE3: Create file studentInformation.txt using vi editor which contains details in the following format.

RegistrationNo:Name:Department:Branch:Section:Sub1:Sub2:Sub3
1234:XYZ:ICT:CCE:A:80:60:70 ... (add at least 10 rows)

- i) Display the number students(only count) belonging to ICT department.
- ii) Replace all occurrences of IT branch with "Information Technology" and save the output to ITStudents.txt
- iii) Display the average marks of student with

ANSWER :

(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2\$ vi studentInformation.txt

```
RegistrationNo:Name:Department:Branch:Section:Sub1:Sub2:Sub3
101:StudentA:ICT:CCE:A:80:60:70
102:StudentB:ICT:IT:B:90:80:85
103:StudentC:CSE:CS:A:70:75:80
104:StudentD:ICT:IT:A:60:65:70
105:StudentE:ICT:CCE:B:85:90:95
106:StudentF:ECE:EC:A:50:60:70
107:StudentG:ICT:IT:B:88:82:90
108:StudentH:ICT:CCE:A:75:75:75
109:StudentI:CSE:CS:B:80:80:80
110:StudentJ:ICT:IT:A:95:90:100
```

(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2\$ vi studentInformation.txt

```
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ grep -c ":ICT:" studentInformation.txt
7
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ sed -e 's/:IT:/:Information Technology:/g' studentInformation.txt > ITStudents.txt
(base) mca@computinglab25-21:~/Desktop/240905110_OSL/Lab_2$ awk -F: 'NR>1 {avg=($6+$7+$8)/3; print $2 " Average: " avg}' studentInformation.txt
StudentA Average: 70
StudentB Average: 85
StudentC Average: 75
StudentD Average: 65
StudentE Average: 90
StudentF Average: 60
StudentG Average: 86.6667
StudentH Average: 75
StudentI Average: 80
StudentJ Average: 95
Average: 0
```